# Jev-Routed Adaptive Temporal Granularity with Recovery and Hysteresis for Efficient Causal Video Generation

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**Abstract**

Adaptive temporal granularity—spending fine per-frame computation only where a causal video segment is hard and cheaper coarse or reused computation elsewhere—promises a better quality–compute frontier than any fixed generation mode. We present a Jev-routed controller that decides, per segment and before generation, whether to run a frame-wise fine mode, a chunk-3 coarse mode, or a near-zero-cost hold mode, stabilized by a hysteresis band and a forced-recovery window. Jev, a text-only System One oracle, predicts segment difficulty, drift risk, and motion class from a BLIP caption of the pre-decision frame, off the diffusion-transformer (DiT) path. On Causal-rCM (Wan2.1-1.3B), evaluated on a Tesla T4 and an A100-80GB, the gains are conditional: under injected $\mathbf{2}\boldsymbol{\sigma_u}$ signal noise the hysteresis-and-recovery band cuts switches by up to **45%** (less on clean traces); on the T4, holding static segments beats fixed fine on the diagnostic reference and cost; and the coarse mode is cheaper only on the A100, where a single-decision Jev policy saves **1.78** s of DiT time and **7.39** s of total time at a small subject-consistency cost. We also audit the three preconditions—cost, signal, and endpoint—showing that a pre-specified hand-crafted signal fails (area under the ROC curve **0.491**) while the Jev oracle scores higher on the same 80 segments (**0.788**).

**Keywords:** video generation, adaptive computation, temporal granularity, hysteresis control, diffusion transformers, efficient inference

# 1 Introduction

Causal video generation with diffusion transformers has become the dominant paradigm for streaming video synthesis and interactive world models, in which frames must be produced one chunk after another under a hard latency budget [6, 21, 36, 37, 39, 41]. Because each chunk is generated conditioned on previously generated context, the rollout invites a natural efficiency idea: not all segments are equally difficult, so one may switch between an expensive *fine* mode (one latent frame per chunk) and a cheaper *coarse* mode (several latent frames per chunk) at run time, deciding from a signal available *before* the segment is generated. This is the video analogue of adaptive computation in recognition networks [4, 26, 28] and of dynamic step allocation in autoregressive video diffusion [9], and it complements work that adapts the video *tokenization* itself [12, 31, 35, 40].

The appeal is a better quality–compute frontier than either fixed mode. The premise, however, rests on three preconditions that are easy to conflate and rarely audited together:

(P1) **Cost.** The coarse mode must be cheaper in wall-clock time at usable quality *on the deployment hardware*—fewer denoising steps do not imply lower latency when each step becomes heavier.

(P2) **Signal.** Some quantity available before generation must identify the segments where coarse generation would hurt.

(P3) **Endpoint.** The quality gain or loss must be certified by an endpoint independent of the modes being compared; an internal reference can silently favor one family.

## 1.1 Research Questions

We organize the audit around four research questions, referred to throughout the paper by number.

**RQ1 (Signal).** Can a quantity measured *before* a segment is generated predict whether coarse generation would be unsafe for that segment, better than chance and better than hand-crafted statistics of the pre-decision context?

**RQ2 (Cost–quality trade-off).** Is there a real wall-clock versus quality trade-off from switching temporal granularity online, and is that trade-off conditional on the deployment hardware?

**RQ3 (Switching control).** Do hysteresis and forced recovery reduce the number of mode switches without worsening rollout regret, and is there a directly measurable quality cost at switch boundaries (a *seam*)?

**RQ4 (Monotonicity).** Is the fine-versus-coarse benefit monotone in the pre-generation signal, so that a threshold on that signal is well posed?

The four questions map onto the three preconditions: RQ1 tests P2, RQ2 tests P1, RQ3 tests the mechanism that stabilizes a controller once P1 and P2 hold, and RQ4 tests whether a scalar threshold is even the right decision rule.

## 1.2 System Overview and Contributions

We audit the preconditions end to end on a real system: Causal-rCM [39], a Wan2.1-T2V-1.3B-480p causal diffusion model [25], using its two shipped chunk families, frame-wise c1-1 and chunk-3 c3-3, plus a near-zero-cost *hold* mode that copies the previous latent frame. Routing is performed per segment by a two-threshold controller with a hysteresis band and a forced recovery window, driven either by hand-crafted signals (residual, optical-flow warp, motion, complexity) or by the *Jev* System One decision model, a text-only oracle that answers typed difficulty, drift-risk, and motion-class questions from a caption of the pre-decision context frame. The audit runs on a

16 GB Tesla T4 and an A100-SXM4-80GB, because the cost precondition is a property of the hardware.

Our findings are deliberately evidence-preserving. On the T4 the coarse family has roughly 2.3× fewer neural function evaluations (NFEs) at the one-step regimes (8.33 vs. 19 averaged over the 45/57/81 grid) but each forward is about 3× heavier, so it is not cheaper; the pre-specified signal test fails outright (held-out AUROC 0.491); and an online warp-threshold policy degenerates to a per-prompt fixed choice at 2.2× the cost of the best fixed policy. On the A100 the cost ordering reverses (coarse is 1.46× faster in DiT time) but the independent endpoint still rejects the hand-crafted policy, because it trades subject consistency for DiT seconds and its own signal decoding erases the total-time gain.

Our contributions are:

1. A paired T4/A100 audit showing that the fine/coarse *cost inversion is hardware-conditional* (per-forward ratio 3.01 vs. 1.60 against a 2.33 break-even point), resolving an apparent contradiction between NFE counts and wall-clock time (Section 6.1, RQ2).
2. A pre-specified (locked-protocol) *negative* result for hand-crafted pre-generation signals on 288 held-out segments, and a higher *same-set* estimate for the Jev System One oracle (AUROC 0.788 $[0.575, 0.946]$ on 80 segments), computed off the DiT path (Section 6.2, RQ1).
3. A recovery-plus-hysteresis controller ablation on both GPUs with a directly measured *seam cost* at switch boundaries (Section 6.7, RQ3).
4. A larger paired A100 evaluation (24 prompts) with prompt-paired bootstrap confidence intervals establishing a *trade-off, not dominance* for Jev-routed generation (Section 6.6, RQ2).
5. An exploratory monotonicity analysis showing that the fine-versus-coarse benefit is monotone in the Jev difficulty signal on the T4 but flat on the A100 online

(diagnostic) split, tying RQ4 to RQ1 and to the choice of endpoint (Section 6.8, RQ4).

We make no claim of universal adaptive superiority; the value is a reproducible account of when the three preconditions hold and what breaks when they do not.

**Relevance to multimedia systems.** Beyond this specific model, the audit speaks to streaming and interactive multimedia: causal video generators are deployed under hard per-segment latency budgets, where the choice of temporal granularity—not just the number of denoising steps—sets the achievable throughput, and where the routing signal must be available before a segment is produced. The findings are directly actionable for multimedia system design: verify the wall-clock cost premise on the target device, prefer a genuinely cheaper reuse mode over a heavier coarse chunk, and certify any saving with an endpoint independent of the modes being compared.

# 2 Related Work

## 2.1 Causal and few-step autoregressive video diffusion

Autoregressive video diffusion replaces bidirectional attention over a full clip with causal attention over generated history, enabling streaming and interactive synthesis. Self Forcing closes the train–test gap by training on the model's own rollouts [6]; Causal Forcing and Causal Forcing++ systematize few-step autoregressive distillation for real-time interactive generation, the latter explicitly contrasting frame-wise and chunk-wise regimes [37, 41]; minWM provides a full-stack open framework [36]; MAGI-1 scales chunk-wise generation [21]. Causal-rCM unifies teacher-forcing consistency distillation with self-forcing refinement and ships frame-wise and chunk-wise students of Wan2.1 [25, 39]; we build directly on its released c1-1 and c3-3 checkpoints. Related designs include Diagonal Distillation [15], Flex-Forcing, which also explores

device-budget-aware inference [16], Next Forcing [32], and semi-autoregressive or non-quantized variants [3, 20]. All fix a single temporal granularity at deployment; we treat granularity as a per-segment decision. Faster autoregressive video diffusion [34] and real-time adversarial post-training [14] established the streaming setting this work builds on.

## 2.2 Adaptive computation and adaptive tokenization

Allocating inference effort by input difficulty is classical: adaptive computation time [4], early exits [26], and learned layer skipping [28]. In video generation, DSA allocates denoising steps dynamically along an autoregressive rollout [9], the closest published point to our fine/coarse regimes, but it does not switch the chunking mode nor study the hardware dependence of the step–latency map. Dynamic computation has also been applied to diffusion transformers for efficient visual generation [38]. A parallel line adapts the tokenizer—temporally causal one-dimensional tokenization [12], adaptive-length tokens [31], dynamic latent frame rate [35], and variable-frame-rate tokenizers [40]—changing granularity at the representation level and offline; we keep the released tokenizer fixed and route among shipped inference modes online.

## 2.3 Error accumulation, surprise, and switching control

Long rollouts accumulate error, and several works respond with content-adaptive mechanisms: entropy-guided guard sampling [5], surprise-based hierarchical chunking [8], surprise-driven memory [23], state-adaptive scheduling [24], and context-aware mixture-of-experts for long horizons [22]. LongScape additionally uses action-guided variable-length chunking in an embodied setting; we instead drive granularity from a pre-generation signal with an explicit recovery window. DASH-OPD introduces discrepancy-aware switching *with hysteresis* for on-policy distillation [29]; our controller reuses the same classical two-threshold band but applies it at inference time to

granularity selection, adds forced recovery, and directly measures the seam cost that hysteresis is meant to amortize.

## 2.4 Caching, memory, and serving efficiency

Complementary savings come from reusing computation rather than choosing modes: content-aware key–value selection [1], flow caching [17], motion-aware caching [33], key–value cache quantization [19], and serving-level resource management [30]. Related efficiency levers include inter-frame denoising reuse [27], content-aware dynamic patchification [11], and data-free quantization of video diffusion transformers [13]. These are orthogonal to granularity routing and composable with it.

## 2.5 Evaluation endpoints

Our independent endpoints use CLIP ViT-B/32 subject consistency and text alignment [18], following the subject-consistency dimension popularized by VBench [7], plus a classical optical-flow suite; context frames are captioned with BLIP [10] for the oracle input. The hardware contrast is mediated by attention kernels—SDPA on the Turing T4 versus FlashAttention on Ampere-class GPUs [2]—which is why the cost precondition is hardware-conditional.

**Positioning.** Adaptive chunk length already exists (Flex-Forcing under a device budget; LongScape for action-conditioned embodied chunking). To our knowledge, however, none decides per segment, online, which temporal granularity to run next from a *pre-generation difficulty signal and* adds an explicit recovery mechanism when that signal fails *and* audits whether the underlying cost premise holds across hardware. That combination is the contribution here.

# 3 Background: Modes and the Operating-Point Trap

Causal-rCM generates a rollout over latent-frame chunks [first_chunk | $\text{chunk}_1$ | $\text{chunk}_2$ | ...], each chunk attending causally to its predecessors [39]. The two shipped families are c1-1 (one initial latent frame, then one-frame chunks) and c3-3 (three initial latent frames, then three-frame chunks), each with documented one-, two-, and four-step midpoint schedules. A $k$-frame chunk mode performs one transformer forward per denoising step *per chunk*, so its NFE count is divided by $k$ relative to the frame-wise family, while each forward processes $k$ times as many latent frames.

This definition creates an *operating-point trap*. An early comparison paired c1-1 in its one-step regime against c3-3 in its four-step regime and found coarse slower in all 72/72 held-out cases (DiT 53.4 s vs. 21.6 s), which naively reads as "coarse is broken" but is in fact a schedule mismatch. The trap generalizes: coarse is cheaper in wall-clock DiT time only if

$$\frac{s_{\text{fwd}}^{\text{c3-3}}}{s_{\text{fwd}}^{\text{c1-1}}} < \frac{\text{NFE}_{\text{c1-1}}}{\text{NFE}_{\text{c3-3}}}, \tag{1}$$

where the right-hand side is set by the chosen schedules. For the one-step regimes used throughout, the schedule break-even is 2.28 from the clean 45/57/81 grid (NFE 19 vs. 8.33) and 2.33 under noisy context (NFE 21 vs. 9); we report 2.33 and note that every conclusion below holds for 2.28 as well, since the measured ratios (2.94–3.01 on the T4, 1.60–1.75 on the A100) lie on the same side of both. The measured per-forward ratio is 3.01 on the T4 but 1.60 on the A100 (Section 6.1): the same model pair lies on opposite sides of (1) on different GPUs, because the Ampere-class A100 with FlashAttention [2] amortizes the heavier three-frame forward far better than the Turing T4 with SDPA. Causal-rCM itself reports non-monotone quality versus refinement for frame-wise one- and two-step schedules [39], which additionally motivates testing whether the fine-versus-coarse benefit is monotone in uncertainty (Section 6.8).

# 4 Method: Jev-Routed Granularity Control

## 4.1 Decision structure

Figure 1 summarizes the system. Rollouts use a fixed grid: a shared coarse prefix of three latent frames, followed by $T$ segments of three latent frames each (four segments at a 57-frame horizon). Before each segment $t$, the controller selects an action

$$a_t \in \{\text{fine (c1-1)},\ \text{coarse (c3-3)},\ \text{hold}\}, \tag{2}$$

where hold copies the previous latent frame at near-zero DiT cost. The action conditions on a pre-generation state only; nothing from the not-yet-generated segment is used.

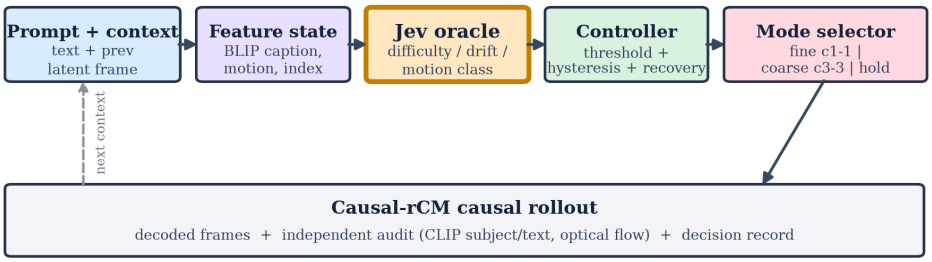



**Fig. 1** Pre-generation routing and causal rollout. For each segment, a feature state (prompt, BLIP caption of the pre-decision context frame, segment index, motion statistics) is sent to the Jev oracle, which returns typed difficulty, drift-risk, and motion-class outputs. The controller state (threshold, hysteresis band, recovery counter) maps these to an action in {fine c1-1, coarse c3-3, hold}, and the selected mode executes the next causal segment. Every decision is recorded with its generated frames, independent quality measures, timing, and switch-boundary cost; all measurements are prompt-paired, and the internal c1-1 reference is diagnostic only.

## 4.2 The Jev System One oracle

Hand-crafted signals from the pre-decision context (residual, warp, motion, complexity; Section 6.2) failed the pre-specified held-out test, so we replace them

with an external *System One* decision model, Jev (TypeSafe API, model identifier `jev-latest`), queried once per decision with a composite typed prompt. The state sent per segment is

$$s_t = (\text{prompt},\ \text{caption}(x_{t-1}),\ t,\ \bar{m}_t,\ \bar{w}_t), \tag{3}$$

where $\text{caption}(x_{t-1})$ is a BLIP caption [10] of the last decoded pre-decision frame and $\bar{m}_t, \bar{w}_t$ are recent motion-magnitude and warp statistics. The oracle returns three typed outputs: a numeric drift risk $r_t \in [0, 1]$ (continuous risk score: whether a coarse chunk would drift or become temporally inconsistent), a difficulty score $d_t$ on a five-level ordinal scale mapped to a number (score type), and a motion class $q_t \in \{\text{static}, \text{moderate}, \text{fast}, \text{scene_change}\}$ (choice type). Mean latency on the T4 audit is 0.672 s per decision ($n$=80), off the DiT path (the input caption is a BLIP decode of a single context frame); the hand-crafted warp signal, in contrast, requires a full VAE decode of the context (≈111 s at 57 frames in the eight-prompt online study; ≈156 s averaged over the 45/57/81 grid). In the A100 online arm the oracle costs ≈1.0 s per rollout for the four-query variants and ≈0.27 s for the single-decision variant (Section 6.6); the per-decision and per-rollout figures come from different hardware and are reported separately. Retries use exponential backoff with a 40 s timeout; the API key is read from the environment and never committed.

### 4.3 Controller: threshold, hysteresis, recovery

Given a scalar signal $u_t$ (the warp statistic or the Jev difficulty score), a threshold controller switches mode when $u_t$ crosses a decision threshold $\tau$. Two mechanisms stabilize it. *Hysteresis* maintains a band $\Lambda = \tau_{\text{high}} - \tau_{\text{low}}$ around the operating point: the mode changes only when the signal exits the band, not on every crossing of $\tau$. *Recovery* forces the fine mode for $R$ consecutive segments after a detected quality

spike (an adjacent-frame $L_1$ jump), then re-arms. Formally,

$$a_t = \begin{cases} \text{fine}, & u_t \geq \tau_{\text{high}} \text{ or recovery active}, \\ \text{coarse/hold}, & u_t \leq \tau_{\text{low}}, \\ a_{t-1}, & \text{otherwise}. \end{cases} \tag{4}$$

Thresholds are calibrated offline from pre-generation data: for the warp controller, $\tau$ is the pooled development median (4.416), the band is $\Lambda = 2\sigma_u$ with $\sigma_u = 3.915$, and $R$=2; for the Jev router, the motion-class rule fitted on the eight-prompt Jev audit set is

$$\begin{gathered} \text{static} \to \text{hold}, \qquad \text{moderate} \to \text{coarse}, \\ \text{fast/scene_change} \to \text{fine}, \end{gathered} \tag{5}$$

with a difficulty-threshold gate ($\tau_{\text{Jev}} = 1.72$, Youden index 0.524) for the difficulty-driven variant. Both the motion-class rule and this gate are fitted on the same eight-prompt Jev audit set whose AUROC we report (Section 6.2); we flag this calibration-on-evaluation-set in Section 9. Over the 32 decisions of the 57-frame subset of that set (8 prompts $\times$ 4 segments), rule (5) assigns 16 coarse, 9 hold, and 7 fine actions—the modes are genuinely mixed, not degenerate. The full Jev audit uses 80 segments (8 prompts $\times$ (4 segments at 57 frames + 6 at 81 frames)).

Two structural properties motivate the design. *Oscillation bound:* for additive signal noise $|\epsilon| \leq \sigma$ with band $\Lambda$, a two-threshold controller's per-step switching probability is bounded by $\Pr[|u+\epsilon-\tau| \geq \Lambda/2]$, so $\Lambda \geq 2\sigma$ eliminates noise-only crossings. The noisy-arm ablation injects $2\sigma_u$ with $\Lambda = 2\sigma_u$, i.e. exactly at the bound's edge rather than strictly inside it (Section 6.7). *Fixed-mode suboptimality:* if coarse attains lower distance than fine on safe segments and vice versa on unsafe ones, the per-segment oracle beats either fixed policy; we record honestly that this proposition's parameter regime was not reachable in deployment, so it is motivation, not a guarantee.

# 5 Experimental Setup

**Model and software.** Causal-rCM Wan2.1-T2V-1.3B-480p [25, 39], checkpoints `c1-1_step2`, `c1-1_step4`, `c3-3_step2`, `c3-3_step2_noisy_ctx`, `c3-3_step4`; umT5-XXL text encoding and the Wan2.1 VAE; guidance 5.0, shift 8.0, 480×832 at 16 fps. Inference is fp16 with the verified single-file recipe (patched dtype, local tokenizer, DiT freed before VAE decode, `expandable_segments`); PyTorch SDPA attention on the T4, FlashAttention-2 on the A100 [2]. All A100 runs pin upstream commit `ed3cb14dd936f92cdc9f9381af7369991509b41f`.

**Hardware.** A Kaggle Tesla T4 (16 GB) and a Google Colab A100-SXM4-80GB. The two machines bracket the deployment regimes of interest: a memory- and throughput-limited Turing card versus an Ampere datacenter card.

**Endpoints.** (i) *Internal diagnostic:* decoded-pixel $L_1$ distance (refL1) to a c1-1 four-step reference at matched prompt, seed, and noise. Because the reference is itself from the fine family, this endpoint can flatter either mode and is never used alone. (ii) *Independent:* CLIP ViT-B/32 subject consistency and text alignment [7, 18]. (iii) *Classical temporal suite:* optical-flow backward-warp residual, frame-difference $L_1$, flow magnitude, and histogram correlation.

**Prompts and splits.** Three development prompts drive the mode sweeps; the pre-specified signal test locks 12 calibration and 24 held-out prompts with seeds 17/29/41 (72 held-out cases, 288 scored segments); eight fresh VBench-derived prompts [7] spanning object, scene, motion, and human-action strata drive the T4 online, controller, monotonicity, and Jev studies; the A100 arm uses fresh manifests of 12 prompts (online policies, controller ablation) and 24 fresh prompts (the larger paired evaluation). Development, calibration, and evaluation splits are disjoint.

**Protocols and statistics.** The master protocol locks the reference construction, the 45/57/81-frame segment grids, the distance fallback, the calibration split, matched-budget baselines (fixed, random-50, alternating-50), grouped prompt×seed bootstrap

intervals, and the preservation of negative results. The locked adaptive-selection criterion requires an adaptive policy to beat fixed fine *and* random same-budget on the independent endpoint at lower measured DiT cost with grouped intervals excluding zero. The A100 protocol adds the hold and latent-refresh policies and a 10,000-resample prompt-paired bootstrap against fixed fine.

# 6 Results

## 6.1 Mode cost is hardware-conditional (RQ2)

Table 1 and Fig. 2(a) give the T4 sweep. The c3-3 family has roughly 2.3× fewer NFEs, but each forward is about 3× heavier (1.96 vs. 0.65 s per forward at the one-step regimes), so coarse is *not* cheaper in wall-clock DiT time, and it is consistently farther from the reference. Fine c1-1 s1 dominates every coarse mode on both axes. Independent endpoints are mixed: coarse modes are less temporally smooth (warp 0.0294–0.0336 vs. 0.0280 for fine s1), while on the three-prompt CLIP sweep the coarse family is not worse (subject consistency 0.9866 for fine s1 vs. 0.9882 for coarse s1; text alignment 0.297 vs. 0.305); fine s1 still dominates on the internal reference and on wall-clock DiT time. A noisy-context variant reduces DiT time for *both* families (within the noisy-context run, fine 30.5 → 15.7 s, −49%, and coarse 35.0 → 21.0 s, −40%; against the clean sweep of Table 1 the reductions are −38% and −29%), so it creates no coarse advantage; we report it as a replicated upstream efficiency trick, not as our novelty.

The same sweep on the A100 reverses the cost clause (Table 2, Fig. 2(b)). The coarse/fine per-forward ratio drops from 3.01 (T4) to 1.60 (timing-only) and 1.75 (real-decode arm), below the 2.33 break-even point of (1): coarse c3-3 s1 is 1.46× faster in DiT time than fine s1 (2.69 vs. 3.92 s). Quality moves the other way: refL1 0.2575 vs. 0.1769, CLIP subject consistency 0.9880 vs. 0.9890, temporal warp 0.0329 vs. 0.0226 (per-mode endpoint values in Table 4). The sweep's internal refL1 favors fine here (fine 0.1769 < coarse 0.2575), but the A100 online arm of Table 8 reverses it (fixed

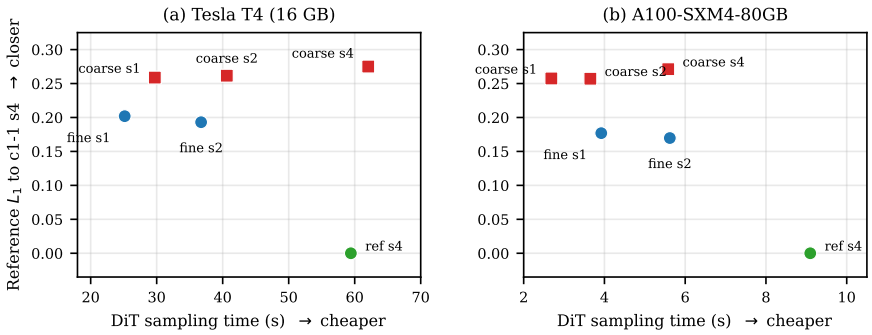



**Fig. 2** Six-mode cost–quality sweep on (a) the Tesla T4 and (b) the A100-80GB. Fewer NFEs do not imply lower DiT time: on the T4 every coarse mode sits above and to the right of fine c1-1 s1, while on the A100 the coarse family occupies a cheaper, less accurate frontier point. The internal refL1 axis is diagnostic (the reference is a fine-family rollout).

**Table 1** Six-mode sweep on the Tesla T4 (45/57/81-frame grid, 3 development prompts, 39 scored pairs). DiT is the mode-dependent cost; VAE decode is mode-independent. refL1 is the internal diagnostic distance to c1-1 s4. Total exceeds DiT+VAE by a mode-independent overhead (text encoding and per-step bookkeeping; ≈12.9 s here and ≈6.6 s on the A100), so the mode-dependent component is DiT time alone.

| Mode | DiT (s) | VAE (s) | Total (s) | NFE | refL1 |
|---|---|---|---|---|---|
| ref c1-1 s4 | 59.42 | 155.7 | 227.8 | 64.0 | 0.0000 |
| fine c1-1 s1 | **25.16** | 156.5 | **194.5** | 19.0 | **0.2019** |
| fine c1-1 s2 | 36.73 | 155.7 | 205.5 | 34.0 | 0.1930 |
| coarse c3-3 s4 | 62.05 | 155.7 | 232.1 | 21.3 | 0.2750 |
| coarse c3-3 s2 | 40.63 | 155.6 | 210.5 | 12.7 | 0.2615 |
| coarse c3-3 s1 | 29.72 | 155.9 | 197.6 | **8.3** | 0.2588 |

fine 0.2832 > fixed coarse 0.2728): the two use different prompt sets and frame grids (the sweep scores 3 development prompts on the 45/57/81 grid; the online arm scores 12 fresh prompts at 57 frames), so the internal endpoint is not comparable across them—the reason it is diagnostic only. VAE decode falls from ≈156 s (T4) to ≈2.4 s and stops dominating the total. *The quality–cost trade-off exists on the A100 and is absent on the T4*: precondition P1 is a property of the GPU. Fig. 3 summarizes the rollout latencies and per-forward ratios against the break-even point.

**Table 2** Six-mode sweep on the A100-80GB with real umT5 encoding and VAE decode (9 cases). The cost ordering of Table 1 reverses; coarse remains less accurate on every quality measure.

| Mode | DiT (s) | VAE (s) | Total (s) | NFE | refL1 |
|---|---|---|---|---|---|
| ref c1-1 s4 | 9.10 | 2.52 | 18.40 | 64.0 | 0.0000 |
| fine c1-1 s1 | 3.92 | 2.38 | 12.95 | 19.0 | **0.1769** |
| fine c1-1 s2 | 5.62 | 2.38 | 14.68 | 34.0 | 0.1697 |
| coarse c3-3 s4 | 5.58 | 2.39 | 14.71 | 21.3 | 0.2712 |
| coarse c3-3 s2 | 3.65 | 2.39 | 12.59 | 12.7 | 0.2571 |
| coarse c3-3 s1 | **2.69** | 2.38 | **11.66** | **8.3** | 0.2575 |

**Table 3** Cost precondition across hardware and context regime. A *forward* is one transformer evaluation on one latent-frame block, as counted in the audit `forwards` field; this count exceeds NFE (e.g. 39 vs. 21 for fine s1 on the T4) because the shared coarse prefix and the guided pass are counted separately, and the released records do not document the exact relation. Times are DiT seconds divided by forward count; the per-forward ratio is compared with the schedule break-even of (1) (2.33; 2.28 gives the same verdict). The per-forward ratio (3.01 on the T4 timing arm) differs from the per-NFE ratio derivable from Table 1 ($\approx$2.69) because forwards $\neq$ NFE. Coarse is cheaper only on the A100, where the ratio falls below break-even.

| Setting | fine s/fwd | coarse s/fwd | ratio | cheaper? |
|---|---|---|---|---|
| T4, timing arm | 0.65 | 1.96 | 3.01 | no |
| T4, noisy context | 0.71 | 2.10 | 2.94 | no |
| A100, timing arm | 0.125 | 0.200 | 1.60 | yes |
| A100, full-decode arm | 0.112 | 0.196 | 1.75 | yes |

**Table 4** Independent-endpoint quality per mode (CLIP ViT-B/32 subject consistency and text alignment; classical temporal-warp mean; $n$=54 videos on the T4, $n$=9 cases on the A100). Lower warp is smoother; higher CLIP is better. On both GPUs the coarse family is less smooth (higher warp) and, on the T4, marginally higher on subject consistency and text alignment.

| GPU | Mode | CLIP subj.↑ | CLIP text↑ | warp↓ |
|---|---|---|---|---|
| T4 | ref c1-1 s4 | 0.9929 | 0.281 | 0.0184 |
| | fine c1-1 s1 | 0.9866 | 0.297 | 0.0280 |
| | fine c1-1 s2 | 0.9894 | 0.295 | 0.0258 |
| | coarse c3-3 s4 | 0.9908 | 0.289 | 0.0294 |
| | coarse c3-3 s2 | 0.9888 | 0.304 | 0.0334 |
| | coarse c3-3 s1 | 0.9882 | 0.305 | 0.0336 |
| A100 | ref c1-1 s4 | 0.9914 | 0.302 | 0.0215 |
| | fine c1-1 s1 | 0.9890 | 0.297 | 0.0226 |
| | fine c1-1 s2 | 0.9907 | 0.298 | 0.0213 |
| | coarse c3-3 s4 | 0.9903 | 0.299 | 0.0262 |
| | coarse c3-3 s2 | 0.9880 | 0.300 | 0.0331 |
| | coarse c3-3 s1 | 0.9880 | 0.301 | 0.0329 |

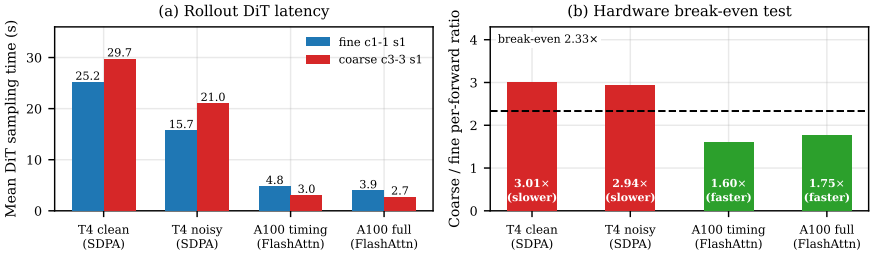



**Fig. 3** Cost precondition across hardware and context regime. (a) Rollout DiT latency for the one-step fine and coarse modes on the T4 (clean and noisy context) and the A100 (timing-only and full-decode arms). (b) The coarse/fine per-forward time ratio against the 2.33 break-even point; the same model pair lies above break-even on the T4 (coarse slower) and below it on the A100 (coarse faster).

### 6.2 The hand-crafted signal fails; Jev is predictive (RQ1)

Table 5 and Fig. 4 report the signal audit. The pre-specified hypothesis fails on locked data: the residual signal's held-out AUROC is 0.491, and its paired advantage over the strongest calibration baseline (warp) is 0.0367 with 95% CI $[-0.0493, 0.1194]$ including zero; the locked complexity signal (0.301, CI $[0.197, 0.412]$) is inverted rather than uninformative, since its interval lies entirely below 0.5. We preserve this as the study's central negative result. On a separate 80-segment T4 audit (label: coarse regret exceeds fine regret; 56/80 positives), the Jev oracle is more predictive: Jev difficulty has the highest point estimate, AUROC 0.788 $[0.575, 0.946]$, versus 0.703 $[0.472, 0.873]$ for its own motion-class output and 0.607 $[0.260, 0.857]$ for the best hand-crafted signal on those same segments. These intervals overlap, and this is a *same-set (calibration-set) estimate*—the motion-class rule and the $\tau_{\text{Jev}}$ gate are fitted on these same eight prompts—so it is not an independent held-out result, and the hand-crafted residual is not scored on these 80 segments. The deployed rule (5) uses the motion class (AUROC 0.703, interval including 0.5), not difficulty; the difficulty-gated variant is in fact the worst Jev policy in Table 8 (total 47.7 s vs. fixed fine 41.8 s; CLIP 0.9545). The oracle answers *before* generation in 0.672 s on the T4, versus a $\approx$111 s VAE decode for the warp signal at 57 frames—about 165$\times$ cheaper and off the DiT path (its only GPU work is a BLIP caption of one context frame). Precondition P2 is therefore plausibly satisfiable, but neither by the hand-crafted suite nor cleanly by an independent estimate.

Figure 5 shows the per-segment decisions of rule (5) on the eight T4 prompts: the traces are non-degenerate and interpretable.

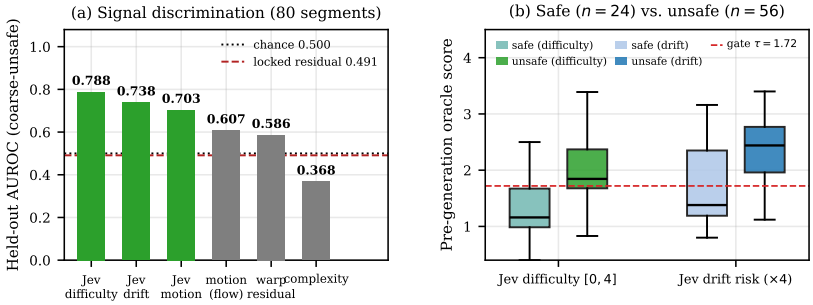



**Fig. 4** Signal discrimination. (a) AUROC on 80 paired T4 segments (coarse-unsafe label; prompt-grouped 95% bootstrap CIs are listed in Table 5): the three Jev outputs have higher point estimates than every hand-crafted signal, but the intervals overlap; the dashed line marks the pre-specified residual result (0.491), and the dotted line is chance. (b) Score separation of the Jev difficulty and drift outputs between coarse-safe ($n$=24) and coarse-unsafe ($n$=56) segments, with the calibrated gate.

**Table 5** AUROC for predicting coarse-unsafe segments. The locked signal study scores 72 held-out cases (288 segments); the residual's held-out value is the study's negative result. The Jev audit uses 80 paired segments from 8 fresh prompts and is a *same-set (calibration-set) estimate* (prompt-grouped 95% CIs, 2,000 resamples).

| Study | Signal | AUROC | 95% CI |
|---|---|---|---|
| Locked | residual | 0.491 | [0.393, 0.589] |
| | warp | 0.454 | [0.343, 0.566] |
| | motion | 0.488 | [0.377, 0.600] |
| | complexity | 0.301 | [0.197, 0.412] |
| Jev audit (80 seg.) | **Jev difficulty** | **0.788** | [0.575, 0.946] |
| | Jev drift risk | 0.738 | [0.479, 0.945] |
| | Jev motion (choice) | 0.703 | [0.472, 0.873] |
| | motion (hand-crafted) | 0.607 | [0.260, 0.857] |
| | warp (hand-crafted) | 0.586 | [0.222, 0.836] |
| | complexity (hand-crafted) | 0.368 | [0.184, 0.562] |

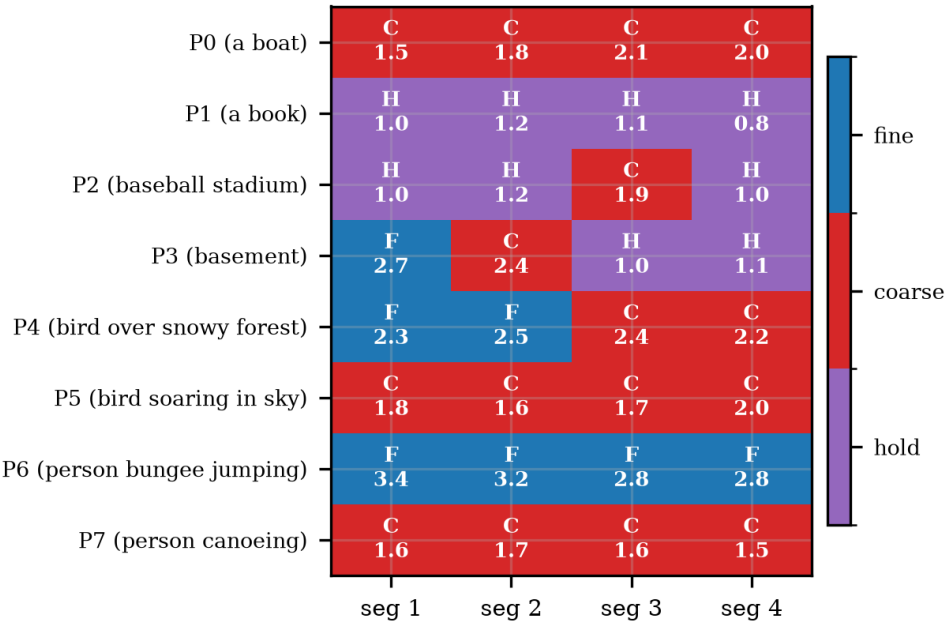



**Fig. 5** Jev routing traces over the eight T4 prompts (57 frames, four segment decisions each). Each cell shows the selected action (F = fine, C = coarse, H = hold) and the oracle's difficulty score. The traces are non-degenerate: static content is held (P1, P2), fast motion is routed to fine (P3, P4, P6), and moderate motion to coarse (P0, P5, P7).

## 6.3 Online routing on the T4: fixed fine dominates

Table 6 and Fig. 6 give the budgeted online test on eight fresh prompts. With no real cost gap on the T4, fixed fine is best on *both* independent quality and cost. The adaptive warp policy gains +1.6% only on the internal reference, is worse on the independent endpoint, costs 2.2× the best fixed policy in total wall-clock (its warp signal forces repeated mid-rollout VAE decodes, inflating VAE time from 110 to 322 s), and makes *zero* within-video switches. Random and alternating same-budget controls sit strictly inside the frontier; the locked adaptive-selection criterion is not met for coarse-mode routing on this GPU.

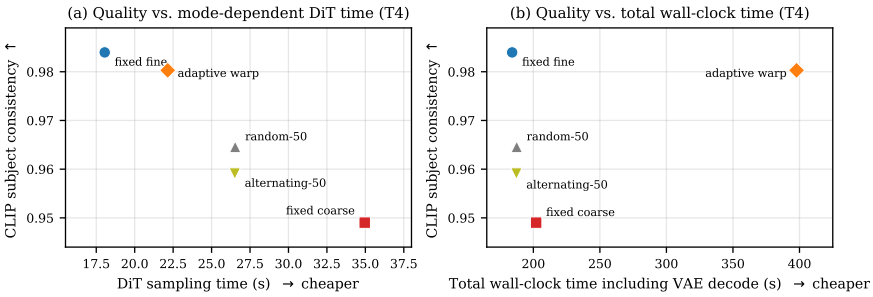



**Fig. 6** Budgeted online policies on the T4 (8 fresh prompts, 57 frames): independent CLIP subject consistency against (a) mode-dependent DiT time and (b) total wall-clock including repeated VAE decodes. Fixed fine dominates on both axes; the adaptive warp policy pays 2.9× VAE cost for its signal and never switches within a video.

**Table 6** Budgeted online policies on the T4 (8 fresh prompts, 57 frames, one shared coarse prefix then four segment decisions). refL1 is the internal diagnostic; CLIP subject consistency and warp are independent endpoints.

| Policy | refL1↓ | DiT (s) | Total (s) | CLIP subj.↑ | warp↓ |
|---|---|---|---|---|---|
| fixed coarse | 0.3186 | 35.0 | 202.1 | 0.9490 | 0.0733 |
| **fixed fine** | **0.2998** | **18.1** | **184.1** | **0.9840** | **0.0303** |
| random-50 | 0.3102 | 26.5 | 187.5 | 0.9646 | 0.0526 |
| alternating-50 | 0.3058 | 26.5 | 187.3 | 0.9591 | 0.0552 |
| adaptive warp | 0.2949 | 22.1 | 397.8 | 0.9803 | 0.0429 |

## 6.4 A hold mode makes routing profitable on the T4

Since the coarse mode is not cheaper on the T4, we add the genuinely cheap *hold* mode and route with rule (5) (Table 7, Fig. 7). Two observations follow. First, routing "moderate" segments to coarse is counterproductive on this GPU—Jev-routed is worse than fixed fine on the internal reference (refL1 0.3085 vs. 0.2998) and on DiT time (24.9 vs. 18.6 s), with total wall-clock roughly equal (182.3 vs. 183.2 s), and its no-hold ablation is worse still—because coarse is slower *and* less accurate there. Second, holding static segments is a better match than re-generating them: Jev-holdfine (static→hold, else→fine) beats fixed fine on the internal (diagnostic) reference and on cost, with refL1 −0.0010, total time −16.2 s (−8.8%), and DiT time −3.5 s (−19%), at unchanged VAE cost; no independent CLIP/warp endpoint was measured for this study, so the quality claim here is diagnostic only. The margin is small and the sample is eight prompts, so we label this a weak positive; it matters because the saving comes from a mode that is actually cheaper, not a signal trick.

**Table 7** Hold-mode routing on the T4 (8 prompts, 57 frames, shared coarse prefix; mode counts are mean actions per rollout out of four). Fixed fine and Jev-holdfine are from the holdfine run ($n$=16); the other rows are from the hold-mode run ($n$=40). The two runs differ in measured timing, so their shared policies are not pooled; the refL1 of fixed fine is identical (0.2998) across both.

| Policy | hold | fine | coarse | refL1↓ | DiT (s) | Total (s) |
|---|---|---|---|---|---|---|
| fixed fine | 0.0 | 4.0 | 0.0 | 0.2998 | 18.6 | 183.2 |
| fixed coarse | 0.0 | 0.0 | 4.0 | 0.3186 | 37.7 | 205.5 |
| fixed hold | 4.0 | 0.0 | 0.0 | 0.3108 | 6.5 | 136.4 |
| Jev-routed | 1.1 | 0.9 | 2.0 | 0.3085 | 24.9 | 182.3 |
| Jev-routed no-hold | 0.0 | 0.9 | 3.1 | 0.3131 | 33.7 | 201.4 |
| **Jev-holdfine** | 1.1 | 2.9 | 0.0 | **0.2988** | **15.1** | **167.0** |

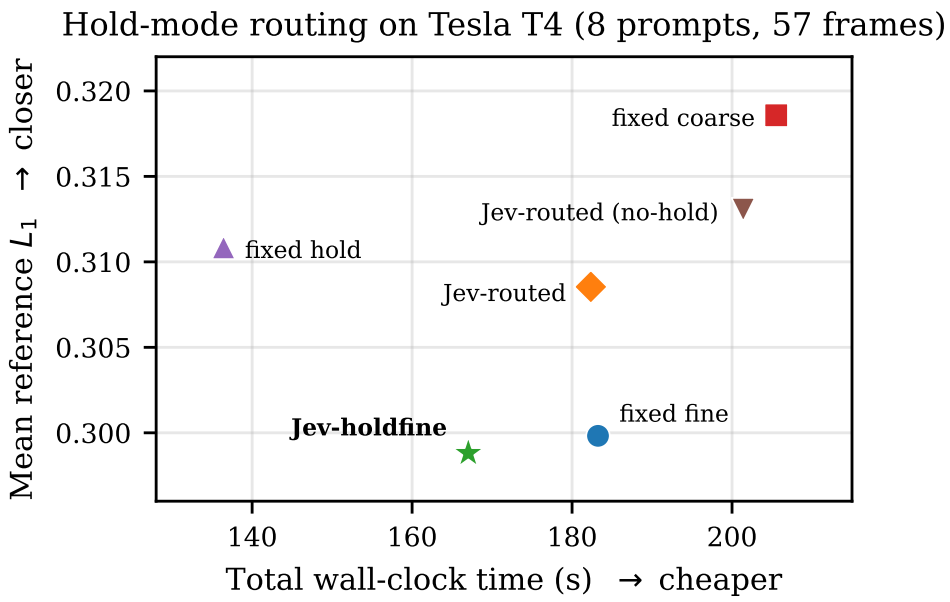



**Fig. 7** Hold-mode study on the T4 (8 prompts, 57 frames; internal refL1 vs. total wall-clock). Routing static segments to the near-zero-cost hold mode (Jev-holdfine, star) beats fixed fine on the internal (diagnostic) reference and on cost; routing moderate segments to the heavier coarse mode (Jev-routed) does not.

### 6.5 The A100 online arm: the endpoint rejects the hand-crafted policy

On the A100 the cost gap is real, so the online test is informative (Table 8, Fig. 8). The adaptive warp policy passes the *internal* gate (+3.5% refL1, −27.9% DiT, beating random same-budget), but the independent endpoint reverses the verdict: CLIP subject consistency 0.9492 vs. 0.9836 for fixed fine, and the signal's extra VAE decodes push total wall-clock above fixed fine (52.3 vs. 46.8 s). On the A100 split the internal reference flatters the *coarse* family (fixed coarse refL1 0.2728 < fixed fine 0.2832), the opposite direction from the T4: a diagnostic endpoint cannot be trusted in either direction, and precondition P3 binds on both GPUs.

The Jev policies, queried in-loop, behave differently. All Jev variants beat the random same-budget control; Jev-holdfine nearly matches fixed fine on the independent endpoint (0.9796 vs. 0.9774) at −35.5% DiT and −16.6% total, and Jev-routed reaches −49% DiT at a larger quality cost. On both A100 online arms, however, *fixed hold* attains the best refL1 (0.2674), the lowest DiT and total time (0.77/12.5 s left; 0.76/11.5 s right), and the highest CLIP subject consistency (0.9926 left; 0.9900 right): because CLIP subject consistency rewards static content, the independent endpoint cannot rank hold-containing policies, and it should not be read as certifying the hold mode. The oracle itself costs ≈1.0 s per rollout including retries, with BLIP captioning at 0.18–0.70 s off the DiT path. A pilot then shows the *single-decision* variant (one oracle query per prompt, reused across segments) cuts in-loop Jev time from 0.97 s to 0.24 s per rollout, while a free latent-space proxy signal is rejected (CLIP 0.9804 vs. 0.9929).

**Table 8** A100 online arms (57 frames, shared coarse prefix). Left: hand-crafted policy set (12 prompts, manifest a100-01); the adaptive warp policy passes the internal gate but fails the independent endpoint and the total-time budget. Right: Jev policies with real in-loop oracle calls (12 prompts); Jev time is included in total time. The left and right sets are separate runs with different prompt manifests, so the shared reference policies (fixed fine, fixed hold) differ in timing between them; their refL1 values coincide.

*Hand-crafted policy set* (12 prompts, manifest a100-01)

| Policy | refL1↓ | DiT (s) | Total (s) | CLIP↑ | warp↓ |
|---|---|---|---|---|---|
| fixed coarse | 0.2728 | 2.55 | 45.5 | 0.9430 | 0.0435 |
| fixed fine | 0.2832 | 3.95 | 46.8 | **0.9836** | **0.0217** |
| fixed hold | 0.2674 | **0.77** | **12.5** | 0.9926 | 0.0223 |
| random-50 | 0.2789 | 3.24 | 46.2 | 0.9588 | — |
| adaptive warp | 0.2733 | 2.85 | 52.3 | 0.9492 | 0.0394 |

*Jev policy set* (12 prompts; Jev time included in total)

| Policy | refL1↓ | DiT (s) | Total (s) | CLIP↑ |
|---|---|---|---|---|
| Jev-difficulty | 0.2678 | 3.19 | 47.7 | 0.9545 |
| Jev-routed | 0.2715 | 2.04 | 33.7 | 0.9614 |
| Jev-holdfine | 0.2739 | 2.59 | 34.9 | **0.9796** |
| fixed fine (ref) | 0.2832 | 4.01 | 41.8 | 0.9774 |
| fixed hold (ref) | 0.2674 | 0.76 | 11.5 | 0.9900 |

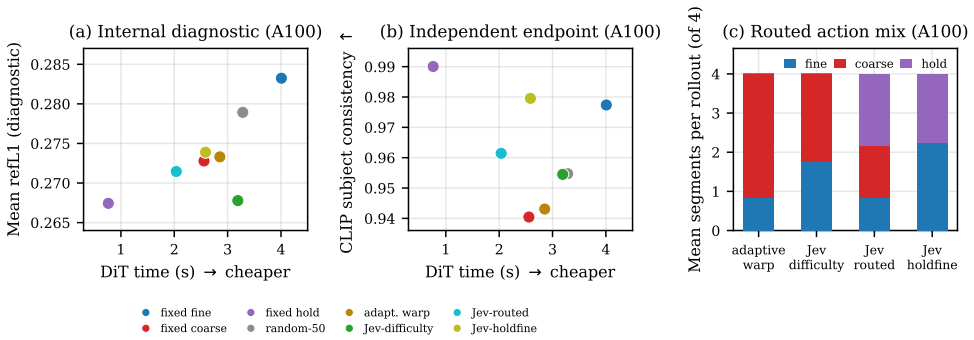



**Fig. 8** A100 online policies (12 fresh prompts, 57 frames) on (a) the internal refL1 diagnostic, (b) the independent CLIP subject-consistency endpoint, and (c) the mean routed action mix per rollout. The internal reference and the independent endpoint rank the same policies differently: the internal reference flatters coarse-family policies, while the independent endpoint rewards the hold-containing policies (fixed hold is highest) because subject consistency rises when static content is frozen.

## 6.6 Larger paired evaluation: a trade-off, not dominance

The larger paired evaluation runs the single-decision Jev policy on 24 fresh prompts with prompt-paired bootstrap intervals (Table 9, Fig. 9). The honest reading is a trade-off: Jev-single saves 1.78 s of DiT time $[-2.20, -1.37]$ and 7.39 s of total time $[-12.39, -3.16]$—both significant—at a small significant loss of $-0.0202$ $[-0.0304, -0.0103]$ in CLIP subject consistency; text alignment is unchanged. The total-time saving exceeds the DiT saving although VAE decode is mode-independent, so the total-time difference partly reflects timing variance and the DiT saving is the mechanistic one. Jev-routed saves the same DiT time without a significant total-time gain. Signal accuracy alone does not certify the system; the independent endpoint is what turns the saving into a measured price.

**Table 9** Larger paired evaluation: prompt-paired bootstrap differences against fixed fine (24 fresh A100 prompts, 10,000 resamples, 95% CIs). Significant intervals are bold. The single-decision Jev policy is a genuine quality–cost trade-off; it is not dominant.

| Policy | ΔCLIP subj. cons. | ΔCLIP text align. | ΔDiT (s) | ΔTotal (s) |
|---|---|---|---|---|
| Jev-single | **−0.0202** [−0.0304, −0.0103] | +0.0013 [−0.0048, +0.0078] | **−1.78** [−2.20, −1.37] | **−7.39** [−12.39, −3.16] |
| Jev-routed | **−0.0180** [−0.0264, −0.0100] | −0.0001 [−0.0045, +0.0055] | **−1.81** [−2.20, −1.41] | −2.58 [−6.62, +1.26] |
| latent refresh | **−0.0131** [−0.0163, −0.0099] | **−0.0037** [−0.0077, −0.0003] | **−0.35** [−0.36, −0.34] | **−0.35** [−0.43, −0.26] |

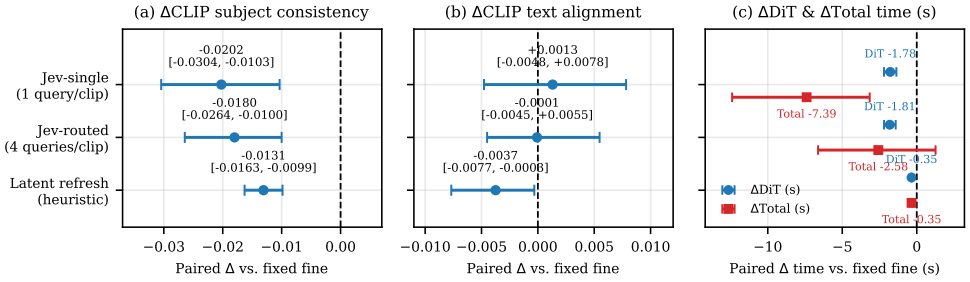



**Fig. 9** Larger paired-evaluation prompt-paired bootstrap differences against fixed fine (24 fresh A100 prompts, 10,000 resamples): (a) CLIP subject consistency, (b) CLIP text alignment, and (c) DiT and total latency. Both Jev policies pay a small, significant subject-consistency cost; text alignment is unchanged; the DiT and total-time savings are significant for the single-decision policy.

## 6.7 Recovery and hysteresis stabilize switching on both GPUs (RQ3)

Table 10 and Fig. 10 ablate the controller on both GPUs with a switch-producing calibration ($\tau$ at the pooled development median, $\Lambda = 2\sigma_u$, $R$=2). Recovery plus hysteresis reduces switching: on the T4, 0.50 → 0.38 mean switches (4→3 events over 8 rollouts) with area-under-horizon refL1 −0.0017 and terminal −0.0067; under injected $2\sigma_u$ noise, 1.12 → 0.62 (9 → 5 events, −45%). On the A100, 0.50 → 0.42 (6 → 5 events over 12 rollouts) on the real signal and 0.92 → 0.50 (11 → 6, −45%) under noise, but the terminal refL1 is marginally *worse* (0.2955 → 0.2977, +0.0022), so the switch reduction is not uniformly free on that GPU. Hysteresis-only and recovery-only are numerically identical to the full controller on these traces (Table 10), so the two components are not separable here. The mechanism's premise—that switching has a

cost—is directly measured: adjacent-frame $L_1$ at switch boundaries exceeds interior positions (0.0364 vs. 0.0280 on the T4; 0.0672 vs. 0.0305 on the A100, 0.0793 under noise), and the full controller reduces the noisy boundary cost (0.0630 $\rightarrow$ 0.0399 T4; 0.0793 $\rightarrow$ 0.0551 A100); the per-arm switch-event counts are small, so the seam figures are descriptive. On the near-monotone real traces the $\Lambda \in \{0.5, 1, 2, 4\}\sigma_u$ and $R \in \{1, 2, 3\}$ sweeps are flat, so the benefit is observable mainly under signal noise. RQ3 is supported on both GPUs for switch reduction, consistent with the oscillation bound.

**Table 10** Controller ablation (RQ3). Switches are mean per rollout (four decisions; T4 $n$=8, A100 $n$=12); absolute switch events: T4 threshold-only 4 $\rightarrow$ 3 (full) and noisy 9 $\rightarrow$ 5; A100 6 $\rightarrow$ 5 and noisy 11 $\rightarrow$ 6. "AUH" is the area-under-horizon *refL1* (the raw diagnostic distance, not a normalised regret). Seam/interior are adjacent-frame $L_1$ at switch boundaries and interior positions (descriptive means; the per-arm switch-event counts are small). Hysteresis-only and recovery-only are numerically identical to full on these traces, so the two components are not separable here. Paired differences are full minus threshold-only.

| GPU | Controller | switches↓ | AUH↓ | terminal↓ | seam / interior |
|---|---|---|---|---|---|
| T4 | threshold-only | 0.50 | 0.3133 | 0.3171 | 0.0364 / 0.0280 |
| | hysteresis-only | 0.38 | 0.3116 | 0.3104 | 0.0321 / 0.0273 |
| | recovery-only | 0.38 | 0.3116 | 0.3104 | 0.0321 / 0.0273 |
| | **full** | **0.38** | **0.3116** | **0.3104** | 0.0321 / 0.0273 |
| | threshold (noisy) | 1.12 | 0.3159 | 0.3172 | 0.0630 / 0.0290 |
| | **full (noisy)** | **0.62** | **0.3150** | **0.3139** | 0.0399 / 0.0260 |
| A100 | threshold-only | 0.50 | 0.2767 | 0.2955 | 0.0672 / 0.0305 |
| | hysteresis-only | 0.42 | 0.2772 | 0.2977 | 0.0601 / 0.0303 |
| | recovery-only | 0.42 | 0.2772 | 0.2977 | 0.0601 / 0.0303 |
| | **full** | **0.42** | 0.2772 | 0.2977 | 0.0601 / 0.0303 |
| | threshold (noisy) | 0.92 | 0.2741 | 0.2997 | 0.0793 / 0.0326 |
| | **full (noisy)** | **0.50** | **0.2718** | **0.2921** | 0.0551 / 0.0315 |

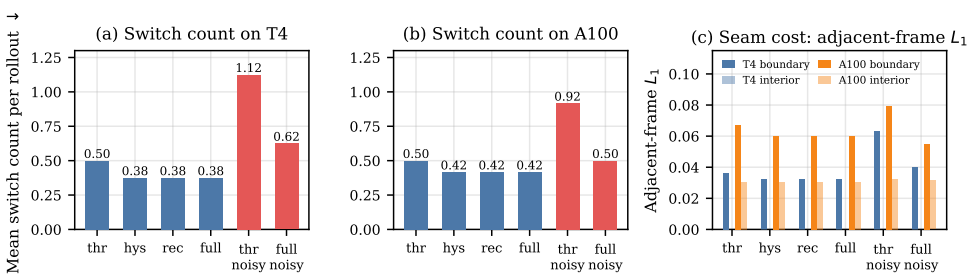



**Fig. 10** Controller ablation (RQ3): mean switch count on (a) the T4 (8 prompts) and (b) the A100 (12 prompts), and (c) the seam cost—adjacent-frame $L_1$ at switch boundaries versus interior positions. Recovery plus hysteresis (*full*) cuts switches on both GPUs, most clearly under injected $2\sigma_u$ signal noise, and reduces the noisy-arm seam cost.

## 6.8 Monotonicity is signal- and endpoint-dependent (RQ4)

The exploratory RQ4 asks whether the fine-versus-coarse benefit is monotone in the pre-generation signal. On 80 paired T4 segments, the warp signal gives $\rho = -0.099$ with non-monotone terciles $(-0.0225, -0.0571, -0.0195)$; the Jev difficulty signal on the same segments flips the result to monotone ($\rho = -0.517$; terciles $+0.0006, -0.0430, -0.0558$; Fig. 11), and the relation holds within both horizons (57 frames: $\rho = -0.458$; 81 frames: $\rho = -0.581$). The earlier non-monotonicity was therefore a *signal-quality artifact*, not a property of the modes. On the A100 online (diagnostic) split, however, the same Jev signal gives $\rho = +0.059$ (AUROC 0.451): no relationship. This is confounded rather than a clean disconfirmation—on that split the internal reference favors coarse (mean fine gap +0.0137; only 38.5% of segments favor fine), so the label carries no stable fine benefit to track. RQ4's verdict is thus *signal- and endpoint-dependent*: monotone with a good oracle against a fine-leaning reference, flat against a coarse-leaning one—the sharpest illustration of why P2 and P3 must be audited together.

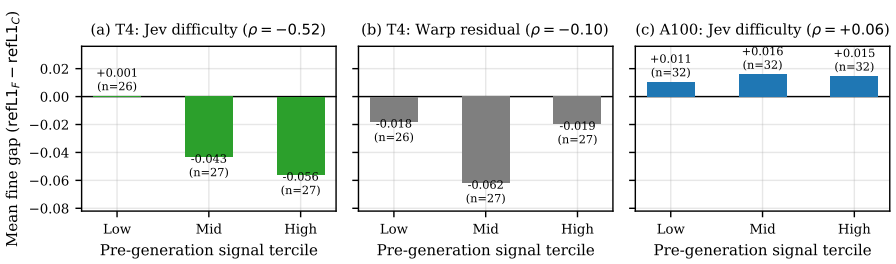



**Fig. 11** Fine-versus-coarse gap across signal terciles. (a) On the T4 with the Jev difficulty signal the fine benefit grows monotonically with difficulty (Spearman $\rho = -0.517$ between difficulty and $\text{refL1}_F - \text{refL1}_C$; negative gap favors fine). (b) The hand-crafted warp signal shows no monotone relationship ($\rho = -0.099$; the mid tercile peaks). (c) On the A100 online (diagnostic) split the same Jev signal is flat ($\rho = +0.059$), because the internal reference there favors the coarse family.

## 6.9 Qualitative evidence

Figure 12 shows matched frames across policies: for a fast-motion human-action prompt the fixed coarse rollout drifts to a different person over a different background while fixed fine and the Jev-routed rollout preserve the subject, and for a bird-over-forest prompt the coarse rollout loses subject identity. We treat these as visual context for the metrics, not as evidence.

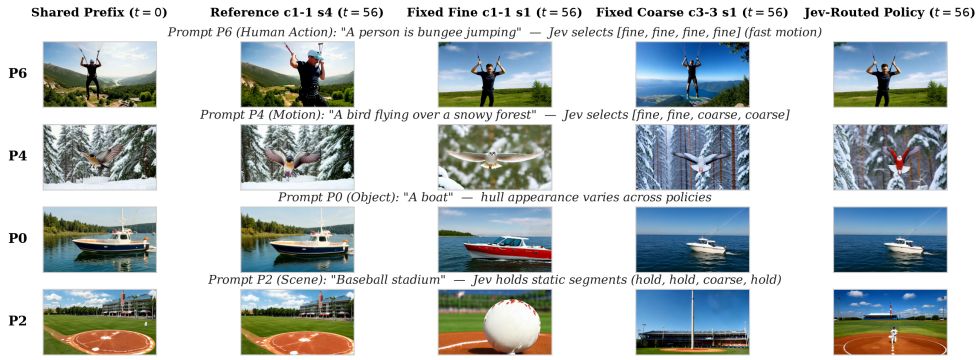



**Fig. 12** Representative decoded frames from matched 57-frame T4 rollouts for four prompts. Each column is the shared prefix frame ($t$=0), the c1-1 four-step reference, fixed fine, fixed coarse, and the Jev-routed policy, all at $t$=56. Row P6 (a person bungee jumping) is routed to fine for all four segments and preserves the subject; row P4 (a bird over snow) mixes fine and coarse and shows coarse identity drift; row P0 (a boat) shows hull-appearance drift across policies; row P2 (a baseball stadium) is held on its static segments. These images are qualitative evidence only; every claim in the paper uses the numeric audits and the independent endpoints.

# 7 Discussion

The audit separates three claims that are easy to merge. *Cost* is a hardware question: whether the c3-3 family's longer temporal stride pays is decided by the per-forward ratio against the schedule break-even (1)—3.01 versus 1.60 here. NFE is not GPU time, and any adaptive-granularity proposal should verify the wall-clock gap on the target GPU before building a controller. *Signal* is a content question: four hand-crafted statistics fail a locked held-out test, while a text-only System One model gives a higher same-set estimate before generation, off the DiT path, at sub-second latency. *Endpoint* is a measurement question: the internal reference reversed its fine/coarse preference across A100 splits (favoring fine in the six-mode sweep, coarse in the online arm), and the A100 online arm shows a policy that "passes" internally while losing subject consistency and total time. Independent endpoints are not optional here.

For practitioners: measure the per-forward ratio against the schedule break-even on the deployment GPU; if no gap exists, route between fine and a genuinely cheap hold mode rather than a heavier coarse chunk; drive the router with an off-GPU oracle; and wrap the switch in hysteresis and recovery.

# 8 Reproducibility

Every quantitative claim resolves to a per-run audit record produced by a named script; the raw rows, generated videos, locked manifests, and calibration freezes are retained with the run inventory, and the complete source tree, figure-generation scripts, and run metadata are released in the public repository **https://github.com/j1s4nn/jev-adaptive-granularity-WM**. Negative results are preserved rather than edited out, so the cost, signal, and endpoint audits can be repeated and independently verified on new hardware.

# 9 Limitations

Several limitations bound the claims. (i) Samples are small and single-seeded: three development prompts for the sweeps, eight fresh prompts for the T4 online, controller, and Jev studies, and twelve and twenty-four fresh prompts for the A100 arms; the results have no confirmatory power for a universal claim, and the paired-evaluation intervals are prompt-paired over $n$=24. (ii) The Jev oracle is external, proprietary, and versioned (`jev-latest`); results depend on its continued behavior, its input is a BLIP caption plus motion statistics of the pre-decision frame (caption noise propagates to the signal), and its AUROC rests on 80 segments from eight prompts with wide intervals; moreover, the motion-class rule and the $\tau_{\text{Jev}}$ gate are fitted on those same eight prompts, so the reported AUROC is a calibration-set estimate rather than an independent held-out one. (iii) RQ2 is a trade-off, not dominance: the paired-evaluation savings and the $-0.020$ subject-consistency loss are both significant. (iv) The internal refL1 reference is a fine-family diagnostic whose direction flips between datasets; the independent CLIP and classical endpoints are the safeguard, but CLIP text alignment on short prompts is noisy, and we report no FVD, PSNR, LPIPS, or full VBench suite because no ground-truth video exists for these rollouts. (v) The study is text-to-video only—no action-conditioned checkpoint was available, so the controller uses no action-conditioned potential, and the EMA uncertainty signal planned in the proposal was unavailable in the verified checkpoints. (vi) Total wall-clock on the T4 is dominated by mode-independent VAE decode; the mode-dependent cost is DiT time, and other backbones or more throughput-bound hardware could shift the ratio. (vii) The fixed-mode-suboptimality proposition's parameter regime was never reachable in deployment; the propositions are retained as motivation.

# 10 Conclusion

Adaptive temporal granularity in a causal video world model is a conditional trade-off whose conditions this paper makes explicit. The cost precondition is a property of the GPU: the chunk-3 coarse mode of Causal-rCM is slower than frame-wise fine generation on a T4 yet 1.46× faster in DiT time on an A100. The signal precondition is a property of the content: a pre-specified hand-crafted suite fails on locked data (AUROC 0.491), while the Jev oracle scores AUROC 0.788 on the same 80 segments in 0.67 s off the DiT path (a calibration-set estimate, not an independent held-out one). The endpoint precondition is a property of the evaluation: the internal reference flatters different families on different splits, and only the independent endpoints turn the A100 savings into a measured, significant price. Within those conditions, the mechanism of the title works conditionally: Jev-routed granularity with a hysteresis band and a recovery window cuts switches by up to 45% under injected signal noise (less on clean traces), routes static segments to a hold mode that beats fixed fine on the internal (diagnostic) reference and on cost on the T4, and delivers a significant DiT and total-time saving at a small significant subject-consistency cost on the A100. Future work should extend the audit to action-conditioned rollouts, larger held-out studies, and blinded human evaluation.

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**Competing interests.** The author declares that he has no competing interests, financial or non-financial, that are directly or indirectly related to the work submitted for publication.

**Data availability.** The complete source tree, locked protocols, calibration freezes, figure-generation scripts, and per-run audit artifacts are openly available in the public

repository at https://github.com/j1s4nn/jev-adaptive-granularity-WM and archived at https://doi.org/10.5281/zenodo.23395414. Large result media (generated videos and figure contact sheets) are hosted at the Kaggle links listed in the repository. Negative and failed experiments are preserved rather than edited out, so that the cost, signal, and endpoint audits can be independently repeated.

**Code availability.** The code and audit artifacts are released under the MIT License at https://github.com/j1s4nn/jev-adaptive-granularity-WM.

**Authors' contributions.** M. J. Hossen conceived the study, designed and ran the experiments, analysed the results, and wrote the manuscript.

**Ethics approval.** Not applicable (no human or animal subjects; the study uses publicly available text-to-video models and text prompts).

**Consent to participate / for publication.** Not applicable.

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**Use of AI tools.** The author used a self-built AI assistant to help collect and organise experimental data, draft and debug code, and draft parts of the manuscript text. All research decisions, the experimental design, the verification of every reported result against the audit records, and the final content of the manuscript were directed, checked, and approved by the author, who takes full responsibility for the work. The Jev oracle is a documented method component (Sections 4.2–4.3), not a writing aid.

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